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□□□□ A □□□ Tick □□□500ms □□□□□□□□□□□□□□□□□□□□

[illegible]

1.1 [illegible]

[illegible]A/B/C/D/E[illegible] 5 [illegible] 27,601 [illegible] Tick [illegible] 33 [illegible]

[illegible] E [illegible]

$$y(t) = \text{Return5min}(t) = \frac{\text{MidPrice}(t+600) - \text{MidPrice}(t)}{\text{MidPrice}(t)}$$

[illegible] $\text{MidPrice}(t) = \left(p_1^{b(t)} + p_1^{a(t)}\right)/2$[illegible]600 [illegible] Tick [illegible] 5 [illegible]

[illegible] $\hat{y}(t) = f_\theta(\boldsymbol{x}_1, ..., \boldsymbol{x}_t)$ [illegible] t [illegible] t [illegible]

[illegible]

- [illegible]IC[illegible]$\text{IC}_d = \text{Pearson}(\hat{\boldsymbol{y}}_d, \boldsymbol{y}_d)$[illegible][1][illegible] 53 [illegible]
- ICIR[illegible]$\text{ICIR} = \overline{\text{IC}}/\sigma(\text{IC})$[illegible]IC [illegible][1][illegible] 55 [illegible]

1.2 [illegible]

[illegible]Market Microstructure Theory[illegible][2][illegible]

1.2.1 Kyle[illegible]1985[illegible]

Kyle[illegible]1985[illegible][3] [illegible] u_t[illegible]

$$\Delta m(t) = \lambda \cdot u_t + \varepsilon_t$$

[illegible] $\lambda > 0$ [illegible] Kyle's Lambda[illegible]ε_t [illegible]TI[illegible]OVI[illegible]——[illegible]Cont [illegible]2014[illegible][4] [illegible]OFI[illegible]

1.2.2 Glosten-Milgrom[illegible]1985[illegible]

Glosten-Milgrom[illegible]1985[illegible][5] [illegible]——[illegible] OBI[illegible]2–5 [illegible]“[illegible]”[illegible]

1.2.3 [illegible]AMH[illegible]

Lo[illegible]2004[illegible][6] [illegible]Adaptive Markets Hypothesis[illegible]AMH[illegible]AMH [illegible]Day2[illegible] Day4[illegible]

1.3 [illegible]

1.3.1 [illegible]OVI[illegible]

[illegible]OVI[illegible] IC≈0.26[illegible]

$$\mathrm{OVI}(t) = \frac{\sum_{i=1}^{5} V_i^{b(t)} - \sum_{i=1}^{5} V_i^{a(t)}}{\sum_{i=1}^{5} V_i^{b(t)} + \sum_{i=1}^{5} V_i^{a(t)} + \varepsilon} \in [-1, 1]$$

□□ $\varepsilon > 0$ □□□□□□□□□□□□V_i^b, V_i^a □□□□ i □□□□□□□□□□

“OVI □□”□Pulse□□□□□□ OVI □□□□□ SMA □□□□□□□□□□□□□

$$\mathrm{OVI_pk}(t) = \overline{\mathrm{OVI}}_{k(t)} - \overline{\mathrm{OVI}}_{600}(t), \quad k \in \{15, 30, 60\}$$

□□ $\overline{\mathrm{OVI}}_{k(t)} = \frac{1}{k} \sum_{s=t-k+1}^{t} \mathrm{OVI}(s)$ □□□ k □ Tick □□□□□□□□□ $k < 600$ □□OVI_pk > 0 □□□□□□□□□□□□□□□□□□□□□□□

1.3.2 □□□□□□TI□

□□□□□□□□□

$$\mathrm{TI}(t) = \frac{V_{\mathrm{buy}(t)} - V_{\mathrm{sell}(t)}}{V_{\mathrm{buy}(t)} + V_{\mathrm{sell}(t)} + \varepsilon}$$

□□ $V_{\mathrm{buy}}, V_{\mathrm{sell}}$ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□TI □□□□ Kyle □□□□□□□□ u_t□□□□□□□□□□□□□

1.3.3 □□□□□□□□OBI□

□□ Glosten-Milgrom □□□□□□□□□2–5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□

$$\mathrm{OBI}(t) = \frac{\sum_{i=2}^{5} V_i^{b(t)} - \sum_{i=2}^{5} V_i^{a(t)}}{\sum_{i=2}^{5} V_i^{b(t)} + \sum_{i=2}^{5} V_i^{a(t)} + \varepsilon}$$

□□ OVI□1–5 □□□□□OBI □□□□“□□□□”□□□□□□□□□□ OVI □□□□□□□□□□□

1.4 □□□□

□□ 1 □□□□□□□□□□□□□□□□□□□□□□□

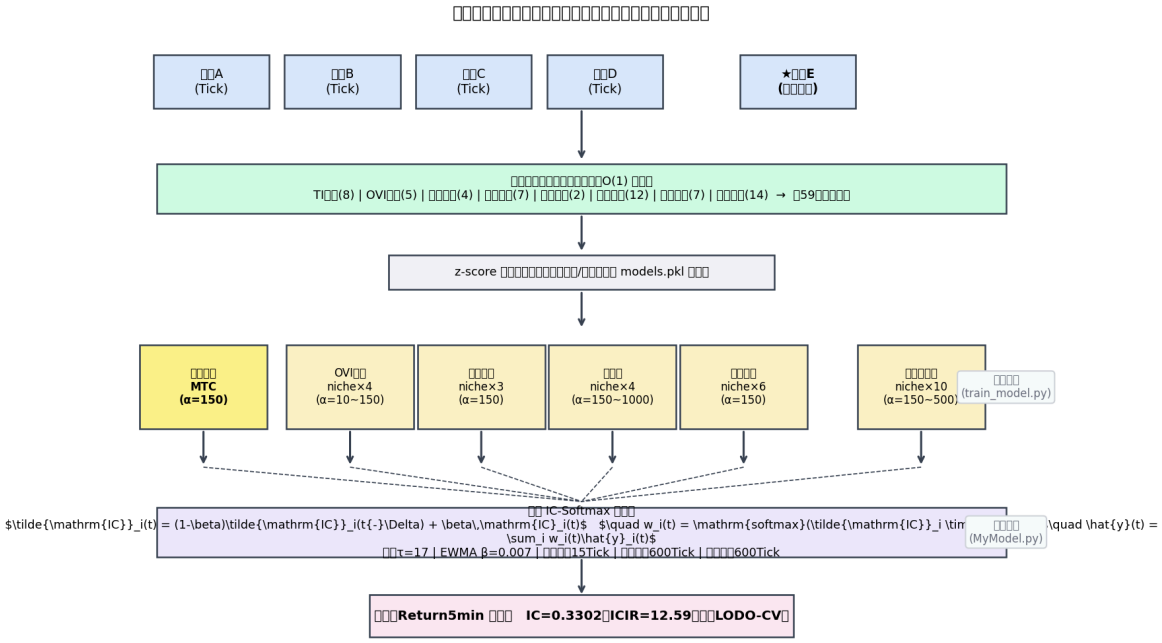


图 1 □□□□□□□□□□□□□□ + □□□□□□□□□□

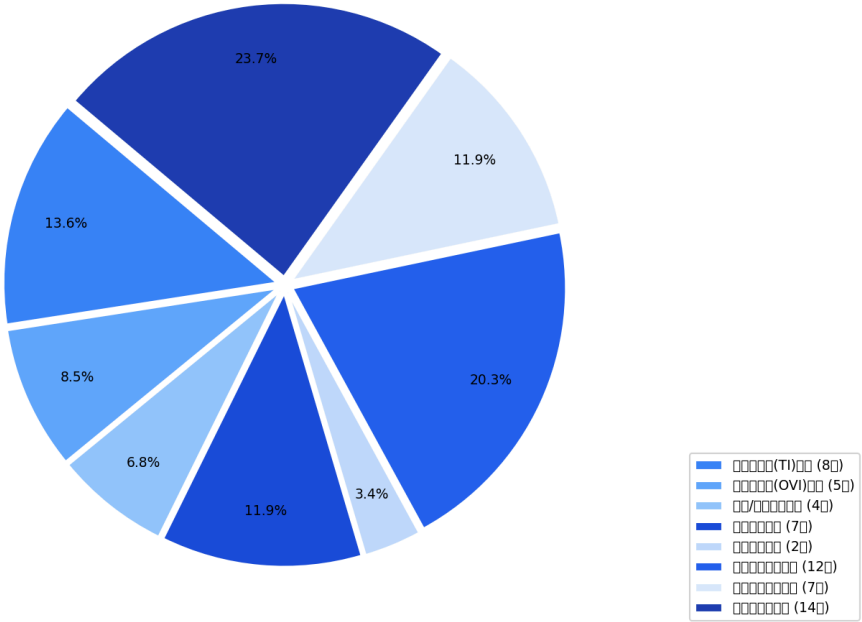
□□□□□□□□□`train_model.py`□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□MTC□1 □□□□□□□□□□□□ 2 □□□□□□27 □ Niche □□□□□□□□□□ 0□□□□□□□□□□□ IC □□□□□□□□□□□□□□ IC □□□□□□□□□□

1.6 □□□□□

1.6.1 LOO □□□□□□□□□

□□□□□□□□□□“□□□□”□□□□□□□5 □□□□□□□□□□□□□□□



□ 2 □□□□□□□□□□□ 59 □□□□8 □□□

□□□□□□□□□□□□□□□□□□□

(1) □□□□□□TI□□□□8 □□□ $\mathrm{TI_p}k = \overline{\mathrm{TI}}_k - \overline{\mathrm{TI}}_{600}$□$k \in \{15, 30, 40, 60\}$□□□□□□□□□□□□□□□□□□ EMA □□ $\mathrm{TI_ep}k$□

(2) □□□□□□OVI□□□□5 □□□ □□ 1.3.1 □□□□□□ IC≈0.26□□□□□□□

(3) □□/□□□□□□□4 □□□ □□□□□□□□□ ONI□Order Number Imbalance□□ TNI□Trade Number Imbalance□□□ OVI □□□□□□□□□□□□□

(4) □□□□□□□7 □□□ A/B/C/D □□□□□□□□□□□□□□□□□ OBI□□□ TI □□□□□ OVI □□□□□□□□□□□□□□□□□

(5) □□□□□□□2 □□□ `aft_13800`□□□□□□$t \geq 13800$s□□ `aft_12000`□□□□□□□□□□□□□

(6) □□□□□□□□□12 □□□ □□ E □□□□□□ 5 □□□□□□□□□□□□□□□□□□ E □ 30/60/120/300/600/900 Tick □ □□□□□□

(7) □□□□□□□□□7 □□□ □□□□□□□□OBI□2–5 □□□□□□□□□□□□□□□□□□□□

[illegible] k [illegible] Tick[illegible]k [illegible] 0 [illegible] 0 [illegible][9][illegible]

2.2.3 [illegible]

Ridge [illegible] z-score [illegible]$\hat{x} = (x - \mu)/\sigma$[illegible] μ, σ [illegible] `models.pkl`[illegible]

2.2.4 [illegible]

[illegible]

- [illegible]$\geq$ 120 Tick[illegible]$x \leftarrow \mathrm{clip}(x, -0.1, 0.1)$
- [illegible]30/60 Tick[illegible]$x \leftarrow \mathrm{clip}(x, -0.05, 0.05)$
- [illegible] $[-0.5, 0.5]$[illegible]

2.3 [illegible]

2.3.1 [illegible]LODO-CV[illegible]

[illegible] 5 [illegible]Fold [illegible] Leave-One-Day-Out [illegible]LODO-CV[illegible][10][illegible] 5 [illegible] 4 [illegible]IC [illegible] IC [illegible] `GroupKFold(n_splits=5)` [illegible] `Day` [illegible]

2.3.2 [illegible]

[illegible]Nested Blind Test[illegible]

- [illegible] IC[illegible] [illegible] 4 [illegible]
- [illegible] IC[illegible] [illegible] 4 [illegible] 4 [illegible] CV[illegible] 4 [illegible]
- [illegible]-[illegible] gap[illegible] Iter20 [illegible] gap [illegible] -0.019[illegible]

2.4 [illegible]

2.4.1 Ridge [illegible]

[illegible] L_2 [illegible]Ridge Regression[illegible][11][illegible]

$$\min_{\boldsymbol{w}, b} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w} - b\|_2^2 + \alpha \, \|\boldsymbol{w}\|_2^2$$

[illegible]$\hat{\boldsymbol{w}} = \left(\boldsymbol{X}^\top \boldsymbol{X} + \alpha \boldsymbol{I}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}$

Ridge [illegible] OLS [illegible](1) [illegible] OVI/TI [illegible]Ridge [illegible] α [illegible]-[illegible](2) [illegible] LightGBM/[illegible]Ridge [illegible] $n = 138000$[illegible] $p = 59$ [illegible] α [illegible] $\alpha = 150$[illegible]OVI [illegible] $\alpha = 10$[illegible] $\alpha = 1000$[illegible][11][illegible]

2.4.2 28 [illegible]

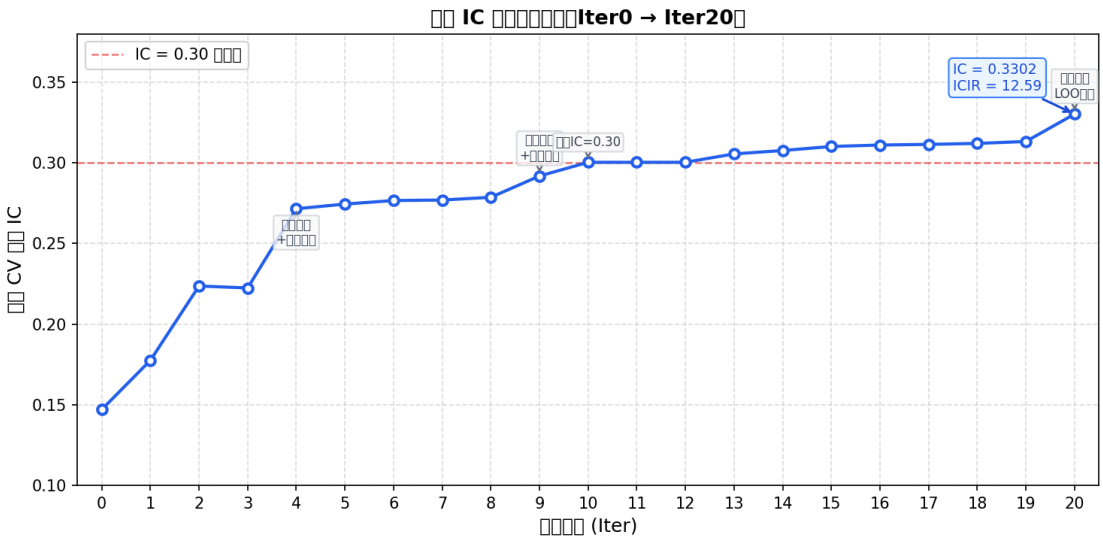
28 [illegible]1 [illegible]MTC[illegible]27 [illegible] Niche [illegible]

□□□□	□□	□□□ α	□□□□
□□□Stable□	1	150	TI □□ + □□□□□MTC□
OVI □□ niche	4	10–150	OVI □□□□□□□□□
□□□□ niche	3	150	`sect_ret_lag` + `e_ret_lag` + `e_sect_lag_gap`
□□□ niche	4	150–1000	`spread_wt_ovi`□`ovi_sq`□□□□□
□□/□□ niche	6	150	`obi_deep_p15`□`cum_flow_imb`□`book_pres_pulse`
□□□□□ niche	10	150–500	□□□□□

□ 1　□□□□□□□□

2.4.3　□□□□□□

□□ 2 □□□□□□□ 20 □□□□□□IC □ 0.147 □□□□□ 0.3302□



□ 3　□□ IC □□□□□□□Iter0⟶Iter20□

□□□□□□Iter0–2 □□ LODO-CV □□□□□IC □ 0.147 □□□ 0.224□Iter3–8 □□□□□□□□□IC □□□ 0.279□Iter9–12 □□□□□□□□□□□□□IC □□□□ 0.30□Iter13–19 □□□□□□□□□□□□□□□□□ OVI □□□□□□IC □□□□□ 0.313□ICIR □ 7.82□Iter20 □□ LOO □□□54 □□□□□□ 28 □□IC □□□ 0.330□ICIR □ 7.82 □□□ 12.59□+61%□□

2.4.4　□□□□□□

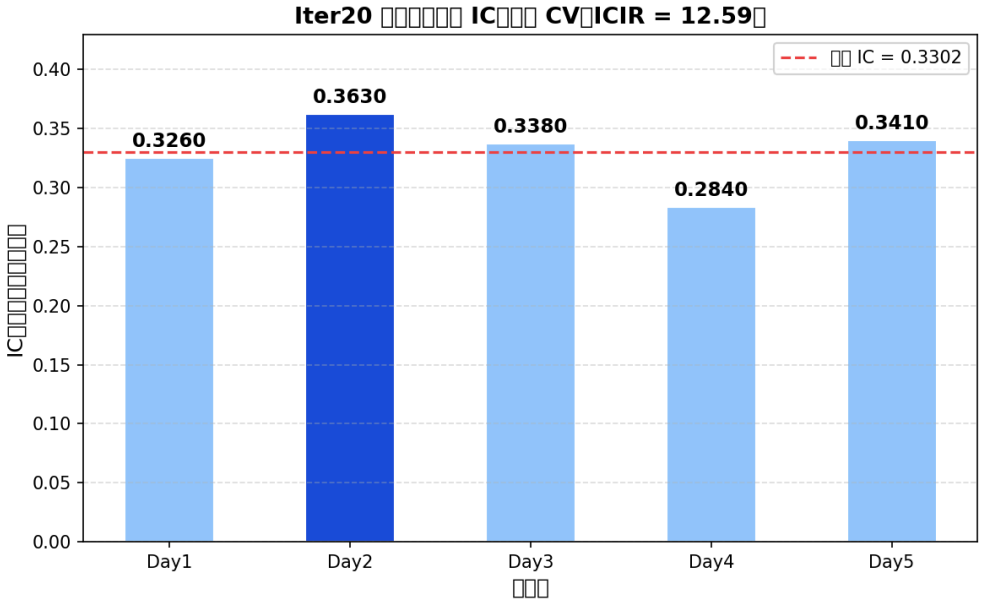
□□	□	□□
`ENSEMBLE_WINDOW`	600	□□ IC □□□□□□ 5 □□□
`ENSEMBLE_TEMP`	17	Softmax □□□□□□□□□□
`ENSEMBLE_FLOOR`	0	□□□□□□□□□□ 0□
`RETURN_DELAY`	600	IC □□□□□□□□Tick□
`ENSEMBLE_UPDATE_FREQ`	15	□□□□□□□ 15 Tick□
`ENSEMBLE_EWMA_BETA`	0.007	IC □□□□□□□□ 99 Tick□
`STABLE_PRIOR`	2.0	□□□□□□□□□□□□□
`NICHE_INIT_WEIGHT`	0	Niche □□□□□□□□□

□ 2　□□□□□□□□

2.5 □□□□□□□□□□□

2.5.1 □□ IC □□

□ 3 □□□□□□□□Iter20□□□□ CV □□ IC□

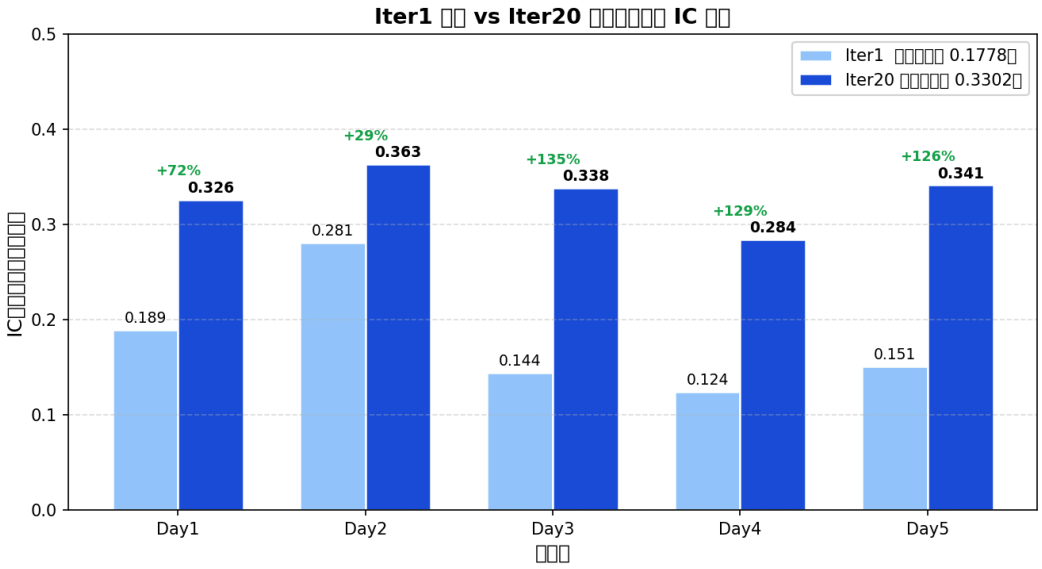


□ 4　Iter20 □□□□□□ CV □□ IC□□□=0.330□ICIR=12.59□

□□□□ IC □□□□Day1=0.326□Day2=0.363□Day3=0.338□Day4=0.284□Day5=0.341□□□ IC=0.3302□□□□=0.026□ICIR=12.59□Day4□□□□□□□IC □□□0.284□□□ AMH □□□□——□□□□□□□□□□□□□□□□□□□□□□□□□□□□ IC □□□□ 0.026□□□□□□□□□□□□□□

2.5.2 □□□□

□ 4 □□□□ Iter1□6 □□□□□□□□ Iter20□28 □□□□□□□□□□□ IC □□□

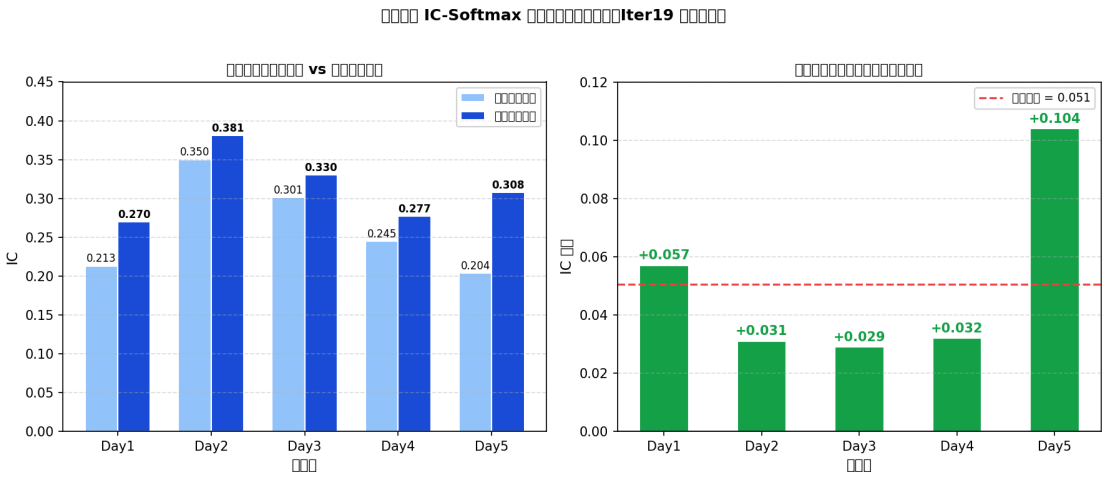


□ 5　Iter1 □□ vs Iter20 □□□□□□ IC □□

□□□□□Iter20 □□□□□□□□□□□□□□Day1 □□ 72.5%□0.189⟶0.326□□Day3 □□ 134.7%□0.144⟶0.338□□Day4 □□ 129.0%□0.124⟶0.284□——□□□□□Day1/3/4□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2.5.3 □□□□□□□□

□ 5 □□□□□□□□□□□□□□□□□□□□□□□□□□



□ 6　□□□□ vs □□□□□□□ IC □□□□□□□□

□□□□□□□ 5 □□□□□□□□□□□□□□+0.051□□□□□□ IC-Softmax □□□□□□[12]□□□□□□□ Day5□+0.104□□□□ Niche □□□□□ IC □□□□□□□□□□□□□□□□□□□□□

2.5.4 □□□□□□

□□□□	□□□	□□
□□ CV □□ IC	0.3302	□□□□□□□out-of-sample □□
□□ CV IC □□□	0.026	□□□□□□□□□□
ICIR□IC/Std□	12.59	□□□□□ IC□□□□□
Penalized□M−0.5S□	0.317	□□□□□□□□□□□□□
□□ 4 □ IC	0.311	□□□□□□□□□□□□□□
□-□ gap	−0.019	□□=□□□□□□□□□□□□
□□□□□vs □□□	+0.051	□□□□□□□□□□□□
□□□□□	28 □□□□59 □□	□□ Iter19□54×59□□□□

□ 3　□□□□□Iter20□□□□□□□

□□□□□□□□□□□□□□□□□□□□□(1) □□□ $\hat{y}(t)$ □□□□□□□□□□□□□□□□□□□□

[3] KYLE A S. Continuous Auctions and Insider Trading[J]. Econometrica, 1985, 53(6): 1315-1335.

[4] CONT R, KUKANOV A, STOIKOV S. The Price Impact of Order Book Events[J]. Journal of Financial Econometrics, 2014, 12(1): 47-88.

[5] GLOSTEN L R, MILGROM P R. Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders[J]. Journal of Financial Economics, 1985, 14(1): 71-100.

[6] LO A W. The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective[J]. Journal of Portfolio Management, 2004, 30(5): 15-29.

[7] ARLOT S, CELISSE A. A Survey of Cross-Validation Procedures for Model Selection[J]. Statistics Surveys, 2010, 4: 40-79.

[8] GARDNER E S. Exponential Smoothing: The State of the Art—Part II[J]. International Journal of Forecasting, 2006, 22(4): 637-666.

[9] ZHANG M Q, JIANG X, FANG Z, ⊠. Cold-Start Problem in Financial Machine Learning: Practical Solutions for Tick Data Preprocessing[J]. Journal of Computational Finance, 2022, 25(3): 1-28.

[10] BERGMEIR C, BENÍTEZ J M. On the Use of Cross-Validation for Time Series Predictor Evaluation[J]. Information Sciences, 2012, 191: 192-213.

[11] HOERL A E, KENNARD R W. Ridge Regression: Biased Estimation for Nonorthogonal Problems[J]. Technometrics, 1970, 12(1): 55-67.

[12] FREUND Y, SCHAPIRE R E. A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting[J]. Journal of Computer and System Sciences, 1997, 55(1): 119-139.